

## SEMESTER 2 EXAMINATION 2025/26

## ADVANCED MACHINE LEARNING

DURATION 120 MINS (2 HOURS)

This paper contains FOUR questions.

Student ID:

Answer all parts of the question in section A (40 marks) and TWO questions from section B (30 marks each)

An outline marking scheme is shown in square brackets to the right of each question.

You **must** write your Student ID in the box and must not write your name anywhere on the paper.

Your answers should be written in the boxes provided on the question paper. The "Additional Space" provided on the facing or adjacent page to the question may be used for corrections or extra space. Clearly indicate which question the answer corresponds to.

Scrap paper can be used - this will be discarded without being looked at.

University approved calculators MAY be used.

## FOR EXAMINER USE ONLY

| Question | Mark | Arithmetic checked | Double Marked |
|----------|------|--------------------|---------------|
| A1       | /40  |                    |               |
| B2       | /30  |                    |               |
| B3       | /30  |                    |               |
| B4       | /30  |                    |               |
| Total:   |      |                    |               |

A foreign language translation dictionary (paper version) is permitted provided it contains no notes, additions or annotations.

**This examination does not contain colour figures.**

**11 page examination paper. This paper has no handouts.**

## Section A

A1

- (a) Explain what *data augmentation* is with a couple of examples of how it is done. Also provide an explanation of what its purpose is. [5 marks]

**Data augmentation is where we manipulate the input data set to increase the effective training set size. The manipulations often reflect some invariance we want to learn. Examples would be to resample an image with a translated bounding box to learn translation invariance. A second example would be to translate a document to a foreign language and back to learn to be invariant to small changes in wording. The purpose is to improve generalisation performance by encouraging the learning machine to learn to be invariant.**

- (b) Provide a brief overview of the *AdaBoost*, explain what task it solves, the structure of the classifier and how it is trained (exact formulae are not necessary). [5 marks]

**AdaBoost performs binary classification. It is constructed as a weighted sum of weak-classifiers,**

$$C_n(x) = \sum_{i=1}^n \alpha_i \hat{h}_i(x)$$

**where  $\alpha_i$  are weights and  $\hat{h}_i(x)$  are weak classifiers with output  $\pm 1$ . The prediction is given by the sign of the strong-classifier  $C(x)$ . The strong classifier is trained greedily adding the weak classifier with the minimum  $\sum_{\text{misclassified } \mu} w_n^\mu$  with  $w_n^\mu = \exp(-y_\mu C_{n-1}(x_\mu))$  (exact equation not necessary). This chooses a weak classifier that tends to misclassify predominantly where the previous strong classifier makes a correct classification. Having chosen the best weak classifier, it learns the weight  $\alpha_n$  (which depends on the weights  $w_n^\mu$  and how  $\hat{h}_n(x^\mu)$ ). The weights  $w_n^\mu$  are multiplicatively updated at each iteration.**

- (c) Explain the kind of problems where the following learning machines excel

- (i) Random Forest
- (ii) Gradient Boosting
- (iii) Support Vector Machines
- (iv) Convolutional Neural Networks
- (v) Gaussian Processes

[5 marks]

**(There are multiple correct answers. Below is an indicative answer only.)**

- (i) Random Forests work well with almost any tabular data where the features can be a mix of categorical and numerical data, with missing data and multiclass problems**

- (ii) Gradient boosting is a regression method. It can handle large data sets where the features can be categorical and numerical
- (iii) Support Vector Machines work well on small dataset of clean data, e.g. all numerical data. Natively they work with binary classification although this can be extended
- (iv) Convolutional Neural Networks work well on data that has local spatial structure which is translationally invariant
- (v) Gaussian Processes is another regression method (although it can be extended to do classification). It works well mainly on numerical data.

(d) Show that for the mapping

$$\mathbf{x} = (x_1, x_2, x_3) \rightarrow \vec{\phi}(\mathbf{x}) = (x_1^2, x_2^2, x_3^2, \sqrt{2} x_1 x_2, \sqrt{2} x_1 x_3, \sqrt{2} x_2 x_3)$$

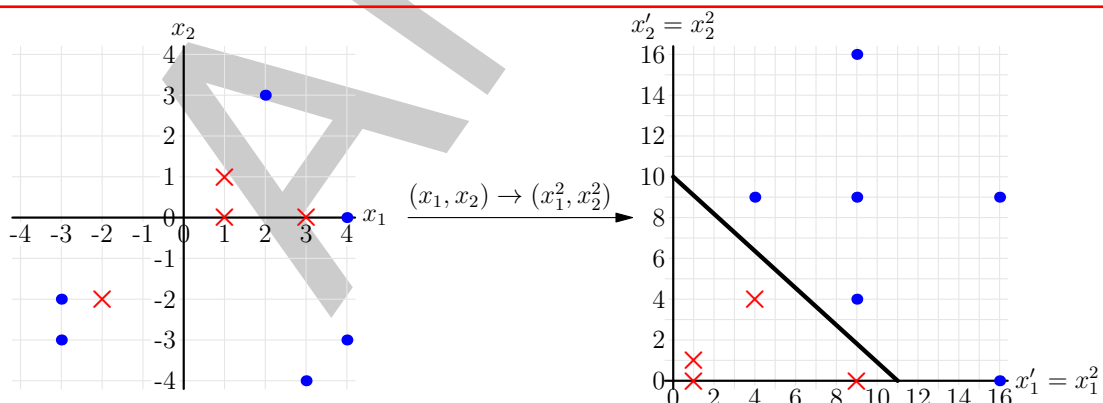
the kernel  $K(\mathbf{x}, \mathbf{y}) = \vec{\phi}(\mathbf{x}) \cdot \vec{\phi}(\mathbf{y})$  is equal to  $(\mathbf{x} \cdot \mathbf{y})^2$ .

[5 marks]

*(This is an extension of the 2-D example shown in the lecturer.)*

$$\begin{aligned}
 K(\mathbf{x}, \mathbf{y}) &= \vec{\phi}(\mathbf{x}) \cdot \vec{\phi}(\mathbf{y}) \\
 &= (x_1^2, x_2^2, x_3^2, \sqrt{2} x_1 x_2, \sqrt{2} x_1 x_3, \sqrt{2} x_2 x_3) \\
 &\quad \cdot (y_1^2, y_2^2, y_3^2, \sqrt{2} y_1 y_2, \sqrt{2} y_1 y_3, \sqrt{2} y_2 y_3) \\
 &= x_1^2 y_1^2 + x_2^2 y_2^2 + x_3^2 y_3^2 + 2 x_1 y_1 x_2 y_2 + 2 x_1 y_1 x_3 y_3 + 2 x_2 y_2 x_3 y_3 \\
 &= (x_1 y_1 + x_2 y_2 + x_3 y_3)^2 = (\mathbf{x} \cdot \mathbf{y})^2
 \end{aligned}$$

(e) In the diagram below show how the datapoints get projected under a projection  $(x_1, x_2) \rightarrow (x_1^2, x_2^2)$ . Sketch the rough position of the separating plane in the extended feature space. [5 marks]



(f) Show that the beta distribution  $\text{Beta}(p|a, b) = p^{a-1} (1-p)^{b-1} / B(a, b)$  is a conjugate prior to the binomial likelihood  $\text{Binom}(k|n, p) = \binom{n}{k} p^k (1-p)^{n-k}$ . Derive update equations for the parameters of the posterior distribution after observing  $k$  successes and  $n - k$  failures.. [5 marks]

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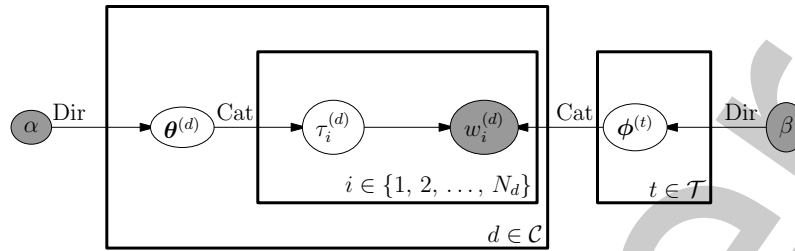
*(Easy if you know what you are doing, but tests real understanding.)*

We only need to consider the functional form with respect to  $p$ . Thus the posterior is proportional to

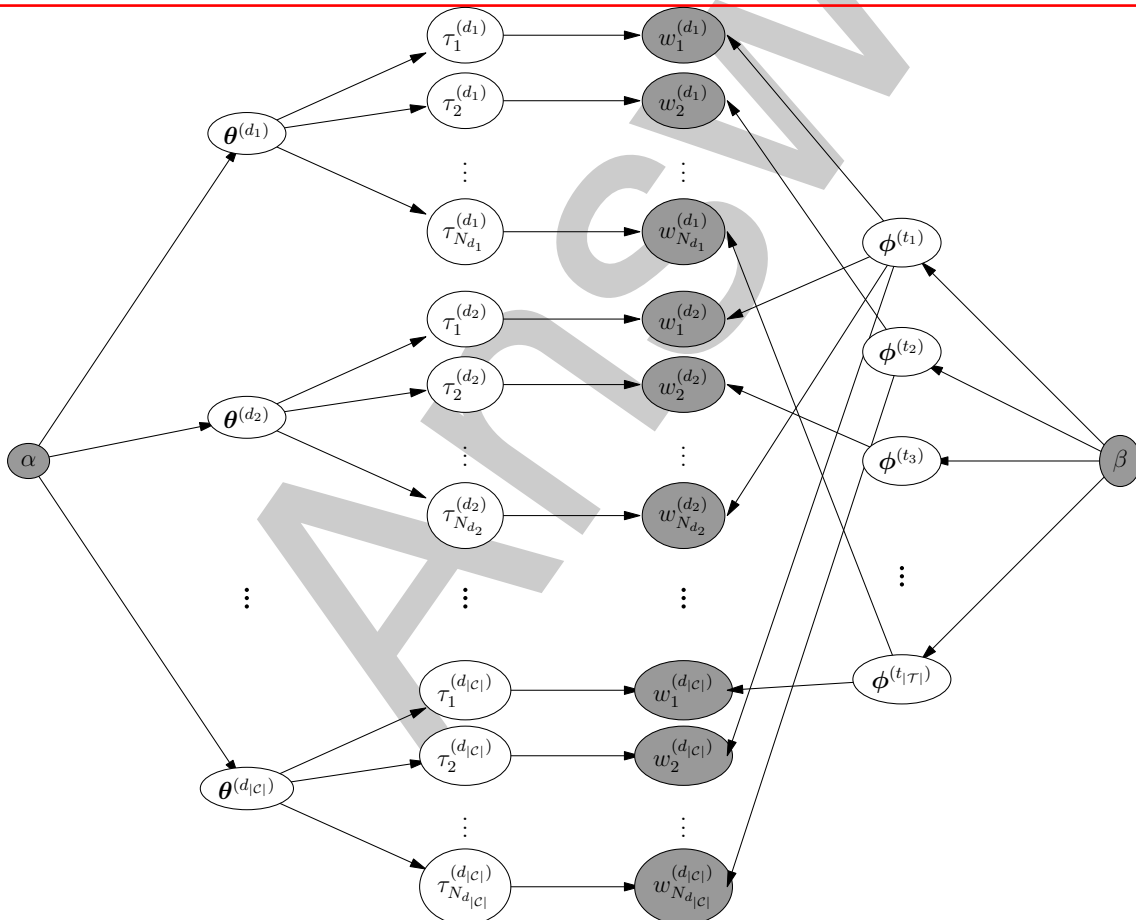
$$f(p|k) \propto p^k (1-p)^{n-k} \times p^{a-1} (1-p)^{b-1} \propto \text{Beta}(p|a+k, b+n-k)$$

The updated equation is thus  $(a, b) \rightarrow (a+k, b+n-k)$ .

- (g) The smoothed latent Dirichlet allocation topic model can be represented as a graphical model by the following plate diagram

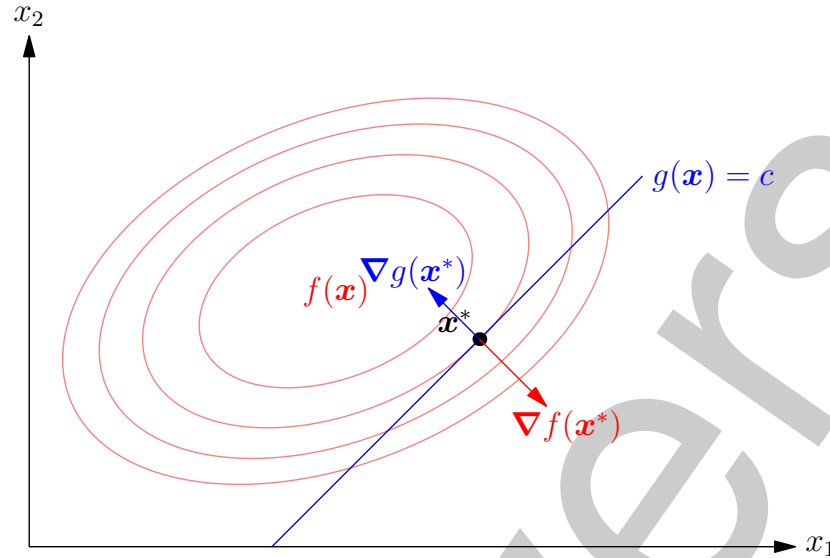


where  $\mathcal{C}$  is a set of documents and  $\mathcal{T}$  is the set of topics. Sketch how documents of size  $N_d$  are generated by expanding the plate diagram to show the full word generation process. [5 marks]



- (h) Below is shown contour lines for a function  $f(\mathbf{x}) = \mathbf{x}^T \mathbf{Q} \mathbf{x}$  that we wish to minimise subject to a linear constraint  $g(\mathbf{x}) = c$ . At the point  $\mathbf{x}^*$  indicated, sketch

the gradient  $\nabla f(x^*)$  and  $\nabla g(x^*)$ . Use the diagram to explain why  $\nabla \mathcal{L} = 0$  finds the solution to the constrained optimisation problem where  $\mathcal{L} = f(x) - \lambda g(x)$ . [5 marks]



(Note the directions are unique up to a change of sign.)

The constraint  $\nabla \mathcal{L} = 0$  picks points where  $\nabla f(x) \propto \nabla g(x)$ . As the gradient is orthogonal to the contour lines this finds points where the constraint (a contour line) is parallel to the contour of  $f(x)$ , which is the condition for a local extremum.

End of question A1

## Section B

B2

(a) Show that the expected generalisation given by

$$\mathbb{E}_{\mathcal{D}}[E(\mathcal{D})] = \mathbb{E}_{\mathcal{D}} \left[ \sum_{x \in \mathcal{X}} p(x) \left( \hat{f}(x|\mathcal{D}) - f(x) \right)^2 \right]$$

can be written as the sum of a bias term,  $B$ , and variance term  $V$  where

$$B = \sum_{x \in \mathcal{X}} p(x) \left( \hat{f}_m(x) - f(x) \right)^2, \quad V = \mathbb{E}_{\mathcal{D}} \left[ \sum_{x \in \mathcal{X}} p(x) \left( \hat{f}(x|\mathcal{D}) - \hat{f}_m(x) \right)^2 \right]$$

where  $\hat{f}_m(x) = \mathbb{E}_{\mathcal{D}}[\hat{f}(x|\mathcal{D})]$  is the prediction made by averaging over all machines.

[15 marks]

**TURN OVER**

*(Students find this derivation difficult because it requires understanding expectations. Being rather conceptual students often struggle with this.)*

The first step is to subtract and add the response of the mean machine

$$\begin{aligned}
 \bar{E}_G &= \mathbb{E}_{\mathcal{D}}[E_G(\mathcal{D})] = \mathbb{E}_{\mathcal{D}} \left[ \sum_{x \in \mathcal{X}} p(x) \left( \hat{f}(x|\mathcal{D}) - f(x) \right)^2 \right] \\
 &= \sum_{x \in \mathcal{X}} p(x) \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(x|\mathcal{D}) - f(x) \right)^2 \right] \\
 &= \sum_{x \in \mathcal{X}} p(x) \mathbb{E}_{\mathcal{D}} \left[ \left( \left( \hat{f}(x|\mathcal{D}) - \hat{f}_m(x) \right) + \left( \hat{f}_m(x) - f(x) \right) \right)^2 \right] \\
 &= \sum_{x \in \mathcal{X}} p(x) \left( \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(x|\mathcal{D}) - \hat{f}_m(x) \right)^2 \right] + \left( \hat{f}_m(x) - f(x) \right)^2 \right. \\
 &\quad \left. + \mathbb{E}_{\mathcal{D}} \left[ 2 \left( \hat{f}(x|\mathcal{D}) - \hat{f}_m(x) \right) \left( \hat{f}_m(x) - f(x) \right) \right] \right)
 \end{aligned}$$

The last term vanishes on taking the expectation. This leaves us with the variance term plus the bias.

(b) Explain the *bias* and the *variance* terms in words. [5 marks]

*(2 marks each and an extra mark if both are correct.)*

- (i) The *bias* is the generalisation performance of the mean machine (i.e. the prediction made by taking the mean response averaged over machines trained with every possible training set of a given size).
- (ii) The *variance* measures how the response of the individual machines vary from machine to machine.

(c) What is the bias-variance dilemma. [5 marks]

To get good generalisation we want to both reduce the bias and variance. To reduce the bias we need a complex machine, but that will typically increase the variance. Conversely a simple machine is likely to have a small variance, but a high bias.

(d) Explain how regularisation helps in the context of the bias-variance dilemma. [5 marks]

By introducing a regularisation term we can use powerful machines (with potentially low bias), however, the regulariser usually makes the learning machine less sensitive to the data, thus reducing the variance.

End of question B2

### B3

(a) Explain for Gaussian Processes (GP) what is the prior, the likelihood and the posterior. [5 marks]

*(Conceptually challenging. This question is from last year, but almost no one tackled it.)*

The prior is a measure over function such that the probability of points in space are normally distributed with a two-point correlation function given by the kernel  $K(x, y)$ —that is it is a Gaussian Process. The likelihood is typically a Gaussian between the observed points and the prediction of the Gaussian process. The posterior is also a Gaussian Process conditioned on the observations.

- (b) Explain what the kernel function represents and how it could be measured empirically from many observations. [5 marks]

The kernel give the two point correlation function. If we have enough data point  $(x_i, y_i)$  we can compute the pairwise distances  $d = \|x_i - x_j\|$ . We can then compute the mean correlations  $(y_i - \mu)(y_j - \mu)/\sigma^2$  for all pairs of points whose distance lies within some narrow interval. Note here  $\mu$  and  $\sigma^2$  are the empirical means and variances of the complete set of target values  $y_i$ . This correlation as a function of the distance function should be similar to the kernel function. Furthermore, within each interval the data should be roughly normally distributed.

- (c) Consider a 1-d Gaussian Process,  $X(t)$ , with a kernel of the form

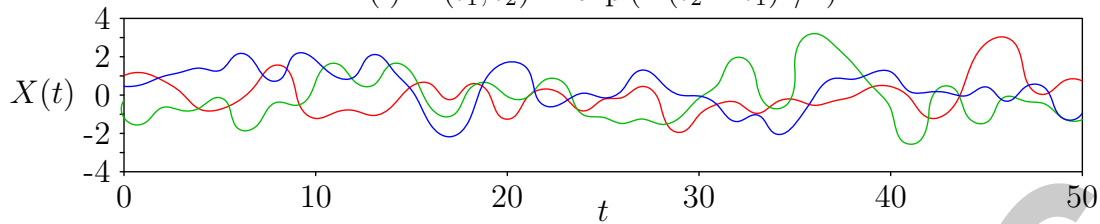
$$K(t_1, t_2) = \exp\left(-\frac{(t_2 - t_1)^2}{2\ell}\right).$$

Sketch three Gaussian Processes drawn from the prior with (i)  $\ell = 1$  and (ii)  $\ell = 2$  (we are not looking for accuracy, but rather the effect of changing  $\ell$ ). [5 marks]

**TURN OVER**

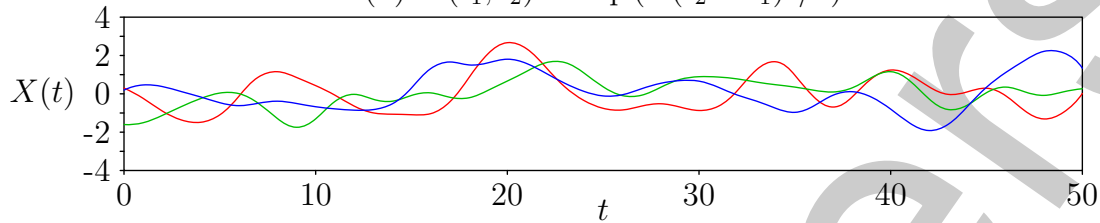
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(i)  $K(t_1, t_2) = \exp(-(t_2 - t_1)^2/2)$



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(ii)  $K(t_1, t_2) = \exp(-(t_2 - t_1)^2/8)$



- (d) Explain the advantages and disadvantage of using the MAP solution rather than a full Bayesian solution. [5 marks]
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The MAP solution is usually much simpler to calculate, particularly when there is no closed form solution for the posterior. Also we don't need to evaluate the normalisation term which often can only be computed approximately with much effort. On the other hand, if the posterior is not unimodal and strongly concentrated around its maximum value then the MAP solution can be very inaccurate. Note also that the MAP solution doesn't give a probabilistic answer so we cannot, for example, compute the accuracy of the prediction (something you can do in the full Bayesian framework).

- (e) Explain why Monte Carlo techniques are often used to solve Bayesian inference problems. [5 marks]
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For many Bayesian problems the posterior has no closed form solution so it cannot be expressed. Monte Carlo techniques allow us to draw samples from the posterior which allow us to compute many quantities of interest such as the posterior mean (often the best single prediction) and the posterior variance (giving an indication of the expected error).

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- (f) Briefly describe in words the use of the MCMC algorithm in Bayesian inference. [5 marks]



In MCMC we explore the posterior distributions by making small jumps in the possible solutions with a probability that satisfies detail balance. That is, the probability of making a move is dependent on the ratio of the posterior probability of the solution before and after the move. This ensures that the distributions of sampled points converges to the posterior distribution. We need to throw away the initial points (burn-in phase). This gives us sample points from the posterior (although to be independent we need to wait for the points to decorrelate).

End of question B3

**B4**

- (a) What guarantees do we have when optimising a convex optimisation problem? [5 marks]

**We are ensured that is a single minimum (which might be a point or a connected region). This is the globally best solution.**

- (b) By considering the Hessian of the loss

$$L(\mathbf{w}) = \frac{1}{2} \mathbf{w}^\top \mathbf{M} \mathbf{w}$$

give the condition on the matrix  $\mathbf{M}$  that must be satisfied for the loss function to be convex. [5 marks]

- (c) By considering the second derivative show that  $f(x) = -\log(x)$  is convex-up. [5 marks]

**The second derivative of  $-\log(x)$  is  $1/x^2$  which is strictly positive. Thus  $-\log(x)$  is convex-up.**

**The Hessian (matrix of second derivatives) is given by  $\mathbf{M}$ . For the loss to be convex the Hessian must be positive definite. Thus we require  $\mathbf{M}$  to be positive definite.**

- (d) Jensen's inequality for convex-up function states that

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Use this to show that for any two categorical distributions  $\mathbb{P}(F = i) = f_i$  and  $\mathbb{P}(G = i) = g_i$ , the KL-divergence

$$\text{KL}(F\|G) = - \sum_i f_i \log\left(\frac{g_i}{f_i}\right)$$

is non-negative.

[5 marks]

**TURN OVER**

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**We note that**

$$\text{KL}(F\|G) = \mathbb{E}_F \left[ -\log \left( \frac{g_i}{f_i} \right) \right] \geq -\log \left( \mathbb{E}_F \left[ \log \left( \frac{g_i}{f_i} \right) \right] \right)$$

**by Jensen's inequality. Writing out the expectation**

$$\text{KL}(F\|G) \geq -\log \left( \sum_i f_i \frac{g_i}{f_i} \right) = -\log \left( \sum_i g_i \right) = \log(1) = 0.$$

- 
- (e) In a variational auto-encoder an input image,  $x$ , is drawn from a dataset  $\mathcal{D}$ . The encoder generates a probability distribution  $q_\phi(z|x)$  defined in a latent space. A vector in the latent space,  $z$ , is sampled from  $q_\phi(z|x)$  and sent to the decoder that then generates a probability distribution in the space of images  $p_\theta(x'|z)$ . The loss function is defined as

$$\mathcal{L}(\phi, \theta) = \mathbb{E}_{x \sim \mathcal{D}} [\text{KL}(q_\phi(z|x) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) - \log(p_\theta(x|z))]$$

Provide an interpretation of the two terms in the loss function. [10 marks]

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**The first term is a KL-divergence or relative entropy between the encoding probability,  $q_\phi(z|x)$ , and a unit normal distribution. It measures the length in nats of the encoding  $q_\phi(z|x)$  where the underlying probability is assumed to be normally distributed. This term punishes distributions that are different from  $\mathcal{N}(\mathbf{0}, \mathbf{I})$  and particularly distributions with very small variance. The second term can be seen as the description length (in nats) of describing the input image  $x$  given the distribution  $p_\theta(\cdot|z)$ . It is small when the probability of sampling  $x$  from  $p_\theta(x|z)$  is large. Minimising the loss finds a compromise between finding an expensive specific code and having to pay for generating images with large errors.**

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End of question B4

**END OF PAPER**

This page contains no examination material